

SP2 Predictions

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Question 1

Consider an ideal/indistinguishable bosonic gas in a constant magnetic field H in a volume V .

Average dipole moment?

Magnetic susceptibility for weak and strong magnetic fields?

Satisfy the Curie law?

Which orientation would BEC occur?

0.0.1 (a)

For bosons we have $m \in \{-1, 0, 1\}$

$$\ln \Xi = - \sum_n \ln \left(1 - e^{-\beta(\epsilon_n - \mu m H)} \right) \approx - \ln \left(1 - e^{-m H \beta \mu} \right) - \int_{-\infty}^{\infty} d\epsilon W(\epsilon) \ln \left(1 - e^{-\beta(\epsilon - m H \mu)} \right) \quad (1)$$

XXX DENSITY OF STATES

$$W(\epsilon) = \delta(\epsilon) + (deg) \frac{V}{h^3} (2\pi) \frac{(2m\epsilon)^{\frac{3}{2}}}{\epsilon} \quad (2)$$

Which gives us in XXX dimensions $\hat{z} = e^{\beta \mu m H + \beta \epsilon(p)}$

$$\ln \Xi = - \ln(1 - z\hat{z}) - \underbrace{\frac{(\deg) V (2\pi) (2m)^{\frac{3}{2}}}{h^3}}_{\alpha} \int_0^{\infty} d\epsilon \epsilon^{1/2} \ln(1 - \hat{z} e^{-\beta \epsilon}) \quad (3)$$

$$\ln \Xi = - \ln(1 - z\hat{z}) + \alpha \left[\frac{2}{3} \epsilon^{\frac{3}{2}} \ln(1 - \hat{z} e^{-\beta \epsilon}) \right]_0^{\infty} - \int_0^{\infty} \frac{2}{3} \epsilon^{\frac{3}{2}} \frac{\hat{z} \beta e^{-\beta \epsilon}}{1 - \hat{z} e^{-\beta \epsilon}} d\epsilon \quad (4)$$

$$\ln \Xi = - \ln(1 - z\hat{z}) + \alpha \left[-\frac{2}{3} \beta^{-\frac{3}{2}} \int_0^{\infty} \frac{(\epsilon \beta)^{\frac{3}{2}} d(\epsilon \beta)}{\frac{e^{\beta \epsilon}}{\hat{z}} - 1} \right] \quad \mathcal{L}i_r(Y) = \frac{1}{\Gamma(r)} \int_0^{\infty} dt \frac{t^{r-1}}{\left(\frac{e^t}{Y} - 1\right)} \quad (5)$$

$$\ln \Xi = - \ln(1 - z\hat{z}) + \alpha \left[-\frac{2}{3} \beta^{-\frac{3}{2}} \mathcal{L}i_{\frac{5}{2}}(z\hat{z}) \right] \text{ explicitly} \quad (6)$$

$$\ln \Xi = - \ln(1 - z\hat{z}) - \frac{2}{3} \beta^{-\frac{3}{2}} \alpha \left[\mathcal{L}i_{\frac{5}{2}}(z) + \mathcal{L}i_{\frac{5}{2}}(ze^{\beta \mu H}) + \mathcal{L}i_{\frac{5}{2}}(ze^{-\beta \mu H}) \right] \quad (7)$$

$$\ln \Xi = - \ln(1 - z\hat{z}) + \frac{V}{\lambda^3} \left[\mathcal{L}i_{\frac{5}{2}}(z) + \mathcal{L}i_{\frac{5}{2}}(ze^{\beta \mu H}) + \mathcal{L}i_{\frac{5}{2}}(ze^{-\beta \mu H}) \right] \quad (8)$$

(9)

The average dipole moment is given by

$$M = \frac{\partial}{\partial H} \ln(1 - z\hat{z}) + kT \frac{\partial \ln \Xi}{\partial H} = -\frac{\beta m \vec{\mu} z \hat{z}}{1 - z\hat{z}} + \frac{kTV}{\lambda^3} \frac{\partial}{\partial H} \mathcal{L}i_{\frac{5}{2}}(ze^{\pm \beta \mu H}) \quad (10)$$

$$= \frac{kTV}{\lambda^3} \left[\mathcal{L}i_{\frac{3}{2}}(ze^{\beta \mu H}) (\pm \beta \mu) e^{\pm \beta \mu H} \right] = \frac{kTV}{\lambda^3} \left[\mathcal{L}i_{\frac{3}{2}}(ze^{\beta \mu H}) (\beta \mu) e^{\beta \mu H} - \mathcal{L}i_{\frac{3}{2}}(ze^{-\beta \mu H}) (\beta \mu) e^{-\beta \mu H} \right] \quad (11)$$

$$M = -\frac{\beta m \vec{\mu} z \hat{z}}{1 - z\hat{z}} + \frac{\mu V}{\lambda^3} \left[\pm \mathcal{L}i_{\frac{3}{2}}(ze^{\pm \beta \mu H}) \right] \quad (12)$$

The magnetic susceptibility is given by

$$\chi = \frac{\partial M}{\partial H} = \frac{\partial}{\partial H} \left(-\frac{\beta m \vec{\mu} z \hat{z}}{1 - z\hat{z}} \right) + \frac{\partial}{\partial H} \frac{\mu V}{\lambda^3} \frac{\partial}{\partial H} \mathcal{L}i_{\frac{3}{2}}(ze^{\pm \beta \mu H}) = \frac{-\beta^2 \vec{\mu}^2 m^2 z \hat{z} (1 + z\hat{z})}{(1 - z\hat{z})^2} \pm \frac{\beta \mu^2 V}{\lambda^3} \mathcal{L}i_{\frac{1}{2}}(ze^{\pm \beta \mu H}) \quad (13)$$

The magnetic susceptibility for weak magnetic fields $\beta\mu H \ll 1$ is given by

$$ze^{\pm\beta\mu H} \rightarrow z(1 + (\pm\beta\mu H) + \dots) \text{ and } \frac{-\beta^2 \vec{\mu}^2 m^2 z \hat{z} (1 + z\hat{z})}{(1 - z\hat{z})^2} \rightarrow \quad (14)$$

$$M \rightarrow \frac{\mu V}{\lambda^3} \left[\mathcal{L}i_{\frac{3}{2}}(z(1 + (\pm\beta\mu H) + \dots)) \right] \quad (15)$$

Taylor exp $\epsilon = \pm\beta\mu H$

$$\mathcal{L}i_v(z(1 + \epsilon)) \approx \mathcal{L}i_v(z) + \epsilon z \frac{d}{dy} \mathcal{L}i_v(y) \Big|_{y=z} + \dots \quad (16)$$

$$\mathcal{L}i_{\frac{3}{2}}(z(1 + (\pm\beta\mu H) + \dots)) \approx \mathcal{L}i_{\frac{3}{2}}(z) \pm \beta H \mu \mathcal{L}i_{\frac{1}{2}}(z) + \dots \quad (17)$$

$$\Rightarrow M \rightarrow \frac{\mu V}{\lambda^3} \left[\left(\mathcal{L}i_{\frac{3}{2}}(z) + \beta H \mu \mathcal{L}i_{\frac{1}{2}}(z) \right) - \left(\mathcal{L}i_{\frac{3}{2}}(z) - \beta H \mu \mathcal{L}i_{\frac{1}{2}}(z) \right) \right] \quad (18)$$

$$M \approx \frac{2H\beta\mu^2 V}{\lambda^3} \mathcal{L}i_{\frac{1}{2}}(z) \quad (19)$$

$$\Rightarrow \chi \approx \frac{2\beta\mu^2 V}{\lambda^3} \mathcal{L}i_{\frac{1}{2}}(z) \quad (20)$$

We see that $\chi \propto \beta \propto \frac{1}{T}$ So it ratifies the curie law.

The magnetic susceptibility for large magnetic fields $\beta\mu H \gg 1$:
we see that $m = 0, -1$ are negligible thus we get

$$\Rightarrow \ln \Xi = \frac{V}{\lambda^3} \mathcal{L}i_{\frac{5}{2}}(ze^{\beta\mu H}) \quad (21)$$

$$M \approx \frac{V\mu}{\lambda^3} \mathcal{L}i_{\frac{3}{2}}(ze^{\beta\mu H}) \quad (22)$$

We observe saturation

$$\chi \rightarrow 0 \text{ as } H \rightarrow \infty \quad (23)$$

For BEC to occur we need an upper bound on the number of particles in the excited states

$$\langle n \rangle = z \frac{\partial \ln \Xi}{\partial z} = z \frac{\partial}{\partial z} \left(-\ln(1 - z\hat{z}) + \frac{V}{\lambda^3} \left[\mathcal{L}i_{\frac{5}{2}}(z) + \mathcal{L}i_{\frac{5}{2}}(ze^{\beta\mu H}) + \mathcal{L}i_{\frac{5}{2}}(ze^{-\beta\mu H}) \right] \right) \quad (24)$$

$$\langle n \rangle = \frac{z\hat{z}}{1 - z\hat{z}} + \frac{V}{\lambda^3} \left[\mathcal{L}i_{\frac{3}{2}}(z) + e^{\beta\vec{\mu}H} \mathcal{L}i_{\frac{3}{2}}(ze^{\beta\vec{\mu}H}) + e^{-\beta\vec{\mu}H} \mathcal{L}i_{\frac{3}{2}}(ze^{-\beta\mu H}) \right] \quad (25)$$